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Review

A review on CRISPR/Cas-based epigenetic regulation in plants

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ABSTRACT

Epigenetic changes are the heritable modifications in genes without altering DNA sequences. The epigenetic changes occur in the plant genomes to regulate gene expression patterns, which were used to regulate different biological processes, including coping various environmental stresses. These changes, including DNA methylation, non-coding RNA regulation, and histone modification, play a vital role in the transcription and translation processes to regulate gene expression. Gene engineering for the development of stress-tolerant crops via the DNA methylation pathway initially needs a proper selection of genes and its promoter. Manipulating epigenetics requires genetic engineering tools such as Zinc finger nucleases (ZFN), transcription activator-like effector nucleases (TALENs), and clustered regularly interspaced short palindromic repeats (CRISPR)/CRISPR-associated protein (Cas). However, CRISPR/Cas9 mediated epigenetic editing refers to transcriptional reprogramming at the targeted sites using epigenetic enzymes fused with decatalytical Cas9 (dCas9). This review focused on the different epigenetic mechanisms in plants and their potential contribution to developing epigenetic tools. The dCas9 endonuclease tethered with transcriptional repressor or activator domain leads to CRISPR inhibitor (CRISPRi) or activator (CRISPRa) for regulating gene expression. The dCas9 has been successfully fused with other various effector domains for constructing epigenetic tools, including the DNA methyltransferase 3A (DNMT3A), or the DNA demethylase TET. Multiple efforts have been made to improve epigenome editing in plants. Initially, incorporating SunTag into the dCas9-EpiEffector complex was used as an epigenetic tool; demethylation of target loci with dCas9-SunTag-TET1 futher increased its efficiency. Additionally, SunTag could also be fused with the dCas9-DNMT3A complex to augment CpG methylation at a targeted loci.

1. Introduction

Epigenetics emerged under genetics, completely associated with genetic materials and phenotype [1]. The heritable alterations, caused by activation or suppression of genes without alteration in the DNA sequences, are known as epigenetic changes. This happens when plants are exposed to various environmental stresses. To cope the climate changes as well as biotic and abiotic stresses, plants need to develop different mechanisms, including epigenetic mechanisms [2]. Various mechanisms inside the plants sensing the environment and enormous changes occur in gene regulation and expression to withstand the multiple stress conditions [3–5]. These epigenetic modifications directly affect plants throughout their lifespan and are inherited in subsequent generations [6,7]. Epigenetic changes cause DNA modifications or structural changes in chromatin rather than DNA sequence changes [6,7]. The epigenetic alterations play a significant role in regulating gene expression by changing chromatin organization and DNA accessibility. Being sessile organisms, plants are inevitably exposed to several abiotic and biotic stresses from the surrounding environment [8,9]. Due to these stresses, several changes occur in plants through biochemical, physiological, morphological conditions that decrease plant growth, photosynthesis activity, and productivity [1,5,8,10–12]. Due to various stresses, there is a decrease in the production of food crops, whereas there is more demand for the growing population. We need to improve crop varieties with desired traits against various stresses to fulfill the entire public demand.

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The development of a new variety of crops can be achieved by traditional breeding and genetic engineering. There are various limitations with conventional breeding, such as interspecies transfer of the desired gene, producing offspring with multiple traits, and time-consuming during backcross. Therefore, genetic engineering is the most promising technology to improve crops. Genetic modification in plants is generally done for the over-expression/down-regulation of the desired genes in crops [13,14]. Transcription, translation, and subsequent protein modification occur once the gene of interest is integrated into the plant genome. The gene expression is controlled by various processes such as DNA methylation, histone modification, and RNA-mediated processes.

Epigenetic engineering is the most promising emerging area in the upcoming days. Engineering plants can be achieved epigenetically through different tools, such as ZFN, TALEN, and CRISPR/Cas9. These tools efficiently modify the plant gene regulation via changes in the promoter region or transcription without altering DNA sequences. The plant epigenetic regulation includes histone-modifying enzymes, non-coding RNAs, and DNA methylation that affect the expressions of various genes. The non-coding RNAs and DNA methylation provide plants with robust epigenetic circuitry. The plant developmental stages are associated with epigenetic regulation, which may promote gene expression modifications and function. Chromosomal changes, gene amplification, and gene duplication, and translocation are also related to epigenetic regulation. Nowadays, plant genome sequencing and genetic screens enable us to understand further the epigenetic regulatory mechanism [15]. Epigenetic modification in plants, such as DNA methylation, histone methylation, acetylation and phosphorylation, and non-coding RNAs, has gained a tremendous attraction for researchers in the past decade. DNA methylation can affect its transcription and functional role of an individual gene. The epigenetic state of chromatin is crucial for successfully expressing the gene and its associated trait [16]. Chromatin modifying enzymes and targeted DNA modification are used to manipulate chromatin and epigenetic mark at the targeted location [17]. Thus, epigenetic modifications are considered as a vital gene regulatory mechanism under severe stress conditions in plants. Stress-responsive genes are controlled by DNA methylation in most plant systems [18,19]. Various plant phenotypes are associated with epigenetic regulation; therefore, study on epigenetic regulation was very important for controlling gene expression for crop improvement [19]. This review discusses various CRISPR/Cas tools for epigenetic modification and their successful application in improving different agronomic traits in plants.

2. Types of epigenetic regulations in plants

There are various types of epigenetic mechanisms, which regulate gene expression without any change in DNA sequences in plants. These include DNA methylation, non-coding RNA regulation, and histone modification; all of them play a vital role in transcription or translation processes [3,4,20]. These epigenetic mechanisms regulate several biological processes, such as the silencing of transposons and transgenes [21]. DNA coded chromatin modifier proteins control the epigenetic changes in organisms [22]. It is also involved in the biogenesis of histone proteins and non-coding RNAs and the remodeling of chromatin in the cell to stabilize the organism's nuclear genome [23,24]. These epigenetic modifications facilitate a cell controling gene expression by specifically turning genes off or on [1]. Accessibility of DNA sequences influences chromatin conformation that leads to coding and regulating DNA sequences, intervening transcription, and finally regulating gene expression in response to various stresses and development [1,25–27].

2.1. Chromatin remodeling

As sessile organisms, plants continuously expose to different abiotic and biotic stresses. These stress conditions severely hamper the plant growth, development, and productivity [28–30]. Plants can sense the environmental changes and respond by initiating the gene expression and maintaining their genome integrity to survive under adverse environmental conditions [29]. The dynamic chromatin surroundings significantly influence gene expression during harsh environmental conditions [28]. Several changes occur during gene expression under adverse environments, which are associated with chromatin modifications [31]. Chromatin modifications are related to the changes in gene expression and are involved in effective mechanisms that facilitate plant growth under stressful conditions [31]. The plants cope with extreme environmental conditions. Plants remember the earlier stress and thus respond more efficiently when they face it again. This kind of counter-strategy is known as priming, associated with chromatin modification, and maybe sustained independently from transcription [32]. Chromatin modifications are heritable epigenetic changes involved in gene expression without changing the nucleotide sequence [33]. Chromatin remodeling, non-coding RNA changes, DNA methylation and demethylation, and histone modifications are crucial in different biological processes [29,34,35]. Chromatin remodeling is involved in several biological functions, including environmental stresses such as biotic stresses (such as pathogen infections) and abiotic stresses (such as drought, salt, and temperature stress) [29]. The chromatin remodeling factors (CHRs) change chromatin's nucleosome position [29]. These CHRs interchange histone modifications, histone variants deposition, DNA methylation, and RNA processing at the post-transcriptional levels [29]. Raxwal et al. observed the increased accessibility of chromatin when *Arabidopsis* was exposed to extreme temperatures, and there was no change in the chromatin accessibility during drought and salt stresses [36]. The histone acetyltransferases (HATs) and histone acetylation and deacetylations (HDACs) exhibited a significant role in cold stress in plants [37]. The HISTONE DEACETYLASE 6 (HDA6) was upregulated during cold stress and enhanced tolerance to freezing conditions in *Arabidopsis* [38]. Under cold stress conditions, the HDACs directly activated DREB1 (dehydration responsive element binding protein1) (ZmDREB1) gene expression and hyperacetylation of histone in maize [39,40].

2.2. DNA methylation and modification

DNA methylation is a preserved epigenetic mark regulating numerous functions, including gene expression, gene imprinting, chromatin structure, and genome stability [41,42]. DNA methylation involves plant response to abiotic and biotic environmental stresses [1,41,42]. Abnormalities in DNA methylation led to plant developmental abnormalities. DNA methylation occurs at CHG, CG, and CHH (where H represents A, T, or C), DNA methylation takes place on both cytosine (5mC) and adenine (6 mA) residue, and it is highly present on transposable elements (TEs) where DNA methylation plays a vital role in silence of their expression at the transcription levels, so-called Transcriptional Gene Silencing (TGS) [43]. The TGS was accelerated when DNA methylation was present at gene regulatory regions. Furthermore, DNA methylation of repeats and intronic TEs has shown influence on mRNA splicing mechanisms [1,41].

2.3. Histone modification

Histone proteins undergo various post-transcriptional modifications (PTMs) such as methylation, acetylation, phosphorylation, sumoylation, and ubiquitination [44,45]. The histone tails are highly basic due to the presence of arginine and lysine. Histone tails wrap the nucleosome core and provide accessibility for DNA-protein and protein-protein interactions [44– 46]. The covalent modifications of histone tails are reversible. The addition and removal of histone marks are catalyzed by various enzymes called writers and erasers, respectively. The proteins, recognizing these modifications (marks), are called readers [45,47]. The histone tails' PTMs change the chromatin structure and facilitate binding

sites of the regulatory proteins [45,47]. The changes of a certain amino acid can also disrupt the stability of the protein structure and further affect protein function.

Methylation of histone tails is documented on arginine and lysine residues by protein arginine methyltransferases (PRMTs) and histone lysine methyltransferases (HKMTs) [48]. A lysine residue can be added one, two, or three methyl groups (me1, me2, and me3) [48], whereas arginine can add a single methyl group [49]. Histone demethylases (HDMs) remove the methyl groups [50]. Lysine-specific demethylase 1 (LSD1) and Jumonji C (JmjC) are two essential enzymes that catalyze the demethylation of amino acid residues in the histone tails [45,47]. Acetylation of histone tails involves the unfolding of the chromatin structure by neutralizing the basic charge of the lysine residues and, as a result lowering DNA binding to histone octamer [45,47]. Histone acetylation and deacetylation of lysines are catalyzed by histone acetyltransferases (HATs) and histone deacetylases (HDACs), respectively [45,47]. Acetyltransferases can add more than one acetyl groups to lysine residues in the histone tails [45,47], involving several abiotic and biotic stress responses in plants [46]. The deacetylation of histone tails acts as a repressor of gene transcription [45,47]. A plant-specific family of TFs (AtHD2C) is regulated by abiotic stress, ethylene, and modulates the gene expression either negatively or positively in Arabidopsis [51].

2.4. RNA modifications

RNAs play different crucial roles in various biological systems, such as messengers (mRNAs), structural components (ribosomal RNAs), translators (tRNAs), regulators (small interfering RNAs), and enhancers (eRNAs) [52–54]. More than 170 different RNA modifications have been identified in plants [53–56]. The chemical modifications of mRNAs, tRNAs, and rRNAs are involved in different functions after co-transcriptional and post-transcriptional modifications of these RNAs [53]. Different types of non-coding RNAs (ncRNAs) like micro RNAs (miRNAs), long non-coding RNAs (lncRNAs), and other small RNAs are also undergoing various types of modifications [57]. They are involved in new functions and interactions with other RNAs in plants [58]. The RNA modifications are essential and widespread molecular mechanisms in plants. The RNA modifications play crucial roles in plant developmental processes, including the development of embryo, shoot meristem, root meristem, leaf initiation, trichome morphogenesis, floral development, and fruit ripening as well as tolerance to different environmental stresses [52,55,59].

The tRNA undergoes around 92 types of chemical modifications to perform a wide variety of functions in plants [60]. The methylation of tRNA guanine and cytosine residues plays a significant role in structural stabilization by binding with Mg^{2+} ions in Arabidopsis [61]. In Arabidopsis, three transfer RNA methyltransferases (TRMs) have been identified, which are involved in the methylation of guanine [62]. Methylation of guanine (*AtTRM11*) showed a significant role in plant growth and development and induced early flowering [62]. The tRNA methylation by a 2′O-ribose methyltransferase in Arabidopsis enhanced immune response to pathogen *Pseudomonas syringae* [63]. The majority of rRNA modifications are regulated by different kinds of protein complexes like small nucleolar RNA, small nucleolar ribonucleoproteins, which target specific sequences of rRNAs [58]. The 25S rRNA methylation was catalyzed by RNA methyltransferases like rRNA cytosine methyltransferase 1 (RCM1) and nucleolar protein 2 (NOP2) in Arabidopsis [64,65]. The Arabidopsis NOP2 encodes three rRNA methyltransferases that are involved in epidermal pattern and root growth [66] as well as chloroplast development under cold conditions [67]. The RNA modifications are performed by special enzymes such as methyltransferases (known as writers), which add methyl groups to adenine (A) and cytosine (C). In contrast, the demethylases (known as erasers) remove methyl groups from the A and C nucleotide bases of RNA. Apart from these two enzymes, another protein molecule involved in the recognition and processing of methylated RNAs was the reader [57,68]. The m^6A (N^6-methyladenosine) methylation of mRNA plays an important role in the adaptation to salt stress in Arabidopsis [69], microspore degeneration in rice [70], and organellar gene expression [71]. The human fat mass and obesity-associated (*FTO*) RNA demethylase increased ~50 % biomass and yield in rice and potato, respectively [72]. The *FTO* (an RNA demethylase) was also involved in several physiological functions such as the proliferation of root meristem, promotion of photosynthetic efficiency, tiller bud formation, and drought tolerance in rice [72].

2.5. Non-coding RNAs

Non-coding RNAs play a pivotal role in several biological functions, involving regulatory mechanisms and acting as main internal signaling modulators [73,74]. The ncRNAs are categorized into small non-coding RNAs (sncRNAs) and long non-coding RNAs (lncRNAs). sncRNAs are RNAs with less than 40 nucleotides in length. sncRNAs are further classified into microRNAs (miRNAs), small nuclear RNAs (snRNAs), small nucleolar RNAs (snoRNAs), signal recognition particle RNAs (srpRNAs), and small interfering RNAs (siRNAs). These all are involved in a divisity of functions in plants [73,75,76]. The RNAs having more than 200 nucleotides long and not translated into protein are known as lncRNAs. LncRNAs are divided as natural antisense transcript (NAT) region, long intronic non-coding RNAs (lincRNAs), long intergenic non-coding RNAs (lincRNAs), and overlapping lncRNAs based on their origin [73,77]. The ncRNAs are involved in multiple functions, such the development of roots, reproductive organs, pollen, fiber development, photomorphogenesis, vernalization [78–83], and involves in response to fungal pathogen infection [84]. lncRNAs form ribonucleoprotein complexes, RNA–DNA hybrids, or RNA–RNA duplexes that are directly linked to their functional activity [42,74,78].

3. Epigenetic regulation in various plants

Recent studies focused on different aspects of epigenetic regulation in plants. The chromatin remodeling, DNA modifications, histone modifications, and RNA modifications are the most constant epigenetic marks to sustain the plants under various stress conditions, such as cold, drought, heat stress, nutrient deficit, heavy metals, and pathogens [85–89]. Abiotic stress and developmental stages are influenced partially by DNA methylation in plants. Among all epigenetic regulation mechanisms, DNA methylation is one of the most important and highly explored in plants. It plays a vital role in epigenetic regulation, such as flowering and hybrid strength [90–93]. Moreover, stress-induced epigenetic alterations are returned to an average level if the stress may be relieved. In contrast, some epigenetic modifications might be stable and inherited to the next generations in plants [47].

3.1. Abiotic stresses

Various environmental stresses, such as Pi (inorganic phosphate) starvation in rice, salt stress in *Arabidopsis*, pesticide (atrazine) in rice, affect DNA methylation [94–96]. Garg et al. [97] observed that DNA methylation influenced the gene expression levels in three different rice cultivars (IR64, Nagina 22, and Pokali) [92]. These three cultivars showed differences in DNA methylation, albeit differently methylated regions associated with varying gene expressions [97]. Under drought stress conditions, increased DNA methylation was observed in the promoter region of *HvCKX 2.1(cytokinin-oxidase 2.1)* in barley. Additionally, plants showed quick shoot induction under drought treatment due to cytokinin's amplitude levels [98]. The drought-induced mutation like modified DNA methylation pattern maintained and inheritable to subsequent generations in rice. Frequent exposure of the plant to drought conditions enhanced the rice plant's adaptability to adverse drought conditions. It indicates that plants' epigenetic regulation may play a vital role in responding to stress conditions [99,100]. A study also showed

that drought stress affected DNA methylation in *Populus* [101]. Heat-induced transcriptional activation with a minor change to a histone modification without a difference in the DNA methylation [102]. Epigenetic regulation of several repetitive elements of Arabidopsis controlled via transcriptional gene silencing leads to prolonged heat stress with short-term heat exposure. The epigenetic modification regulates the heavy metal tolerance and mycorrhizal associations in Poplar [103]. Like histone deacetylase (HDA9), histone modification modulates plant response to drought and salt stress in Arabidopsis [104].

3.2. Biotic stresses

Pathogen infection also affects plant growth and development as well as survival. Four LDL genes like LDL1 (At1g62830), LDL2 (At3g13682), LDL3 (At4g16310), and LDL4 (At3g10390) are responsible for different kinds of physiological functions such as circadian clock regulation, hormone responses, homologous recombination, flowering time control, and systemic resistance in Arabidopsis [105–110]. Methylation of lysine 4 of histone (H3) protein was involved in the plant immunity during the infection of *Pseudomonas syringae* in Arabidopsis [111]. The LDL proteins involved plant immunity in Arabidopsis [111)]. The rice infected with *Xanthomonas oryzae* pv. Oryzae induced methylation (H3K4me2/3) of histone caused disease resistance [112]. The bacterial pathogen, *P. syringae* pv. tomato (Pst) upregulated the defense signal by activating defense and hormone pathways in DNA methylation, which confered resistance to the pathogen in Arabidopsis [113]. DNA methylation involved resistance against downy mildew disease caused by Hyaloperonospora arabidopsidis (Hpa) in Arabidopsis [114]. The DNA hypermethylation enhanced disease resistance to two fungal pathogens like Plectosphaerella cucumerina and Alternaria brassicicola in Arabidopsis [114].

3.3. Plant growth and development

Plant response to various environmental conditions is one of the best examples of epigenetic regulation, such as flowering time in plants. Plants can remember the flowering time that is a long period of exposure to low temperatures in the spring season [115,116]. Moreover, epigenetic mechanisms are helpful to survive the plant in stressful conditions. The plant has stress memory to survive in stressful situations and is inherited to the offspring of stress-treated generation [117,118]. Brzezinka et al. [119] identified the genes that are prerequisites for heat stress memory in Arabidopsis. The short-term temperature stress increased the photosynthetic yield in the progenies of the Arabidopsis [120]. The *FORGETTER1* (*FGT1*) gene was induced by heat stress, mediated by chromatin and nucleosome; it directly binds to heat-inducible genes and requires heat stress memory [121]. In Arabidopsis, the memory of temperature is accessed from their ancestors so that next-generation progeny withstands high-temperature conditions [122–124]. Thus, epigenetic regulation and its hereditary feature are crucial for crop breeding programs for coping environmental stresses [124]. In plants, methylation of H3 lysine 4 is associated with the expression of high temperature-responsive genes [124]. The developmental epigenome is crucial for fruit ripening and maturation [125]. The control and expression of DELAY OF GERMINATION1 regulated by abscisic acid (ABA) signaling pathway encoding a regulator of LDL1 regulate seed dormancy [106]. H3K4me2 recognized the damaged DNA sites during homologous recombination repair, which maintained the integrity and stability of the gene with the help of LDL1 by preventing the dissociation of RAD54 [109]. LDL2 in association with LDL1 regulates the circadian clock and the primary seed dormancy [106–108]. All these findings suggest that epigenetic changes might be helpful for crop improvements through genome editing tools.

4. Epigenetic engineering tools

Genetic and epigenetic engineering requires different tools in plants [1]. The epigenetic modification was a challenge before the genome editing era. Zinc finger nucleases (ZFN) and transcription activator-like effector nucleases (TALENs) in the area of genome engineering revolutionized epi-genome engineering tools. Both ZFN and TALENs are fused to epigenetic modifiers, which enable epigenetic modifications at a specified locus. Epigenome editing was first demonstrated in *Arabidopsis* against to FLOWERING WAGENINGEN (FWA) promoter with ZFN fused to RNA-directed DNA methylation factor SUVH9; this led to FWA silencing with targeted methylation [126,127]. However, designing tools with TALEN and ZFN proteins was tedious, time-consuming, and difficult for multiplexing [127]. Discovery of clustered regularly interspaced short palindromic repeats (CRISPR) and its associated endonuclease (Cas) revolutionized epigenetic modification in an easier way. The CRISPR/Cas mediated genome editing systems include two components, Cas9, gRNA or three components, spacer, direct repeats (or sgRNA scaffolds), and Cas9. CRISPR/Cas9 genome editing technology is widely applied in various plant species [127–129]. CRISPR/Cas9 genome editing tool is easy to design and user-friendly as compared to the existing editing tools. Advancement in CRISPR/Cas9 technology makes multiplexing easier in a single construct to target multiple genes at a time. The development of deactivated Cas9 (dCas9, also called dead Cas9), which can bind at the target region but do not have catalytical activities, makes various other tools [128]. RNA aptamer is an RNA oligonucleotide that binds to a specific target molecule with higher affinity and specificity. RNA aptamers have been used in epigenetic modification tools by fusing with gRNA. MS2 and PP7 are RNA aptamers well-characterized to bind MS2 and PP7 coat protein. The effector molecules fused with MS2 and PP7 coat protein are finally used for epigenetic regulation as well as chromatin imaging. The MS2-CRISPR/dCas9 system has been used for chromatin imaging [130].

4.1. dCas9-SunTag system

The dCas9 fused with a transcriptional activator domain such as VP64, hybrid VP64-p65-Rta (VPR) activators, and transcription activation domain-like synergistic activation mediator (SAM) has been used to increase the transcription of the desired gene. The dCas9-SunTag system is a very strong transcriptional activator that consists of two units: dCas9 fused to tandem GCN4 peptide repeats and a single-chain variable fragment (scFv) GCN4 antibody fused to super folder-GFP (sfGFP) and VP64 (Fig. 1) [129].

4.2. dCas9 fused with DNA methyltransferase (DNMT) or demethylase (TET)

DNA methylation of a gene is related to the post-transcriptional gene silencing, while promoter methylation leads to transcriptional gene silencing [19]. In plants, DNA methylation occurs via RNA-directed DNA methylation [41]. Cytosine methylation occurs at CG, CHG, and CHH nucleotide sequence (H can be adenine, cytosine or thymine). Methylation is controlled by DNMT and TET. Scientists fused the dCas9 with DNMT or TET and used this tool as an epigenetic regulator in plants. TET1 causes demethylation of DNA via oxidation of 5mC to 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC), and 5-carboxylcytosine (5caC). TET is well characterized in humans, whereas it is lacking in plants. A previous study showed that the ectopic expression of the human TET3 catalytic domain in Arabidopsis involved DNA methylation. Using TET1, researchers developed a molecular tool targeting locus-specific DNA demethylation in Arabidopsis [131].

Tobacco DRM methyltransferase catalytic domain fused with dCas9 was used in Arabidopsis for epigenetic modification of FLOWERING WAGENINGEN (FWA) locus [129]. In Arabidopsis, epigenetic modification at the FWA locus in their progeny depends upon the levels of CG

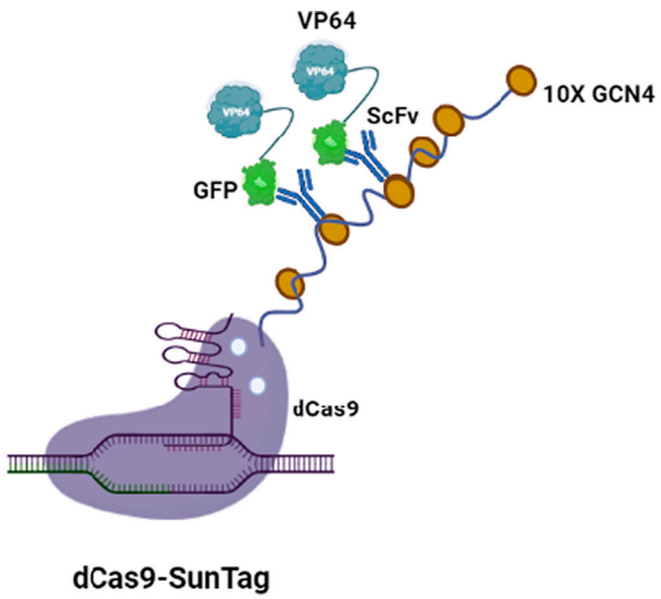

Fig. 1. DeadCas9-SunTag system: 10× GCN4 peptides appended in array to dCas9 carboxy-terminus. 10× GCN4 peptides fused to single-chain variable fragment (scFv) antibodies. scFv has an affinity for multiple GFP and VP64. The figure is drawn by biorender.com.

methylation. The main disadvantage of tobacco DRM domain fusion with scFv and dCas9 is that there is a high level of ectopic CHH methylation among the Arabidopsis genome [129]. This can be overcome by reducing the abundance of methyltransferase into the nucleus, which can be done by removing one NLS in the dCas9-DRM chimeric protein (Fig. 2) [129]. The use of CG-specific bacterial methyltransferase will be more useful in plants, although it is less effective. CG-specific methyltransferase M.SssI with ZFN was also used for methylation of FWA promoter in Arabidopsis. This tool also showed heritable DNA methylation to the FWA promoter and other off-target sites [50]. All these experiments induced successfully epigenome alteration that further caused different trait modifications in plants. Epigenome editing technology is used to develop stress resistance in plants against adverse environmental conditions. Additionally, this epigenetic mark allows stable improvement in agricultural practices with new traits [21].

4.3. p300 acetyltransferase fused dCas9

Addition of the acetyl group to histone tails is called acetylation. Acetylation of histone tails enables the unfolding of the chromatin structure through neutralization of the lysine residues with its basic charge, hence lower availability of DNA binding to histone octamer [45]. Histone acetyltransferases are diverse proteins classified into type A and B. Type A histone acetyltransferases are localized in the nucleus, whereas B is localized in the cytoplasm [132]. The dCas9–p300 acetyltransferase enhances the acetylation of histone H3 lysine 27 (H3K27) near the promoter and enhancer region to increase the gene expression. Histone H3K27 acetylation reduces DNA–histone binding strength and favors DNA accessibility by promoting nucleosomes spacing. H3K27 acetylation in plants used the mammalian p300 histone acetyltransferase fused to dCas9 (Fig. 3). The chimeric dCas9-p300 targeting to FLOWERING TIME (FT) promoter (regulator of flowering time) resulted in increased H3K27 levels in plants. However, increased H3K27 level affected FT transcript level, and further affected the flowering time [128].

4.4. CRISPR/dCas9-LSD

In this tool, these structurally unstable domains increase the fusion protein yield through proteasome-mediated degradation. Small molecules like 4OHT [4-hydroxytamoxifen] can attach and stabilize those domains and stabilize ER. Similarly, binding of TMP [trimethoprim] stabilizes DHFR enables the control of gene expression. Histone

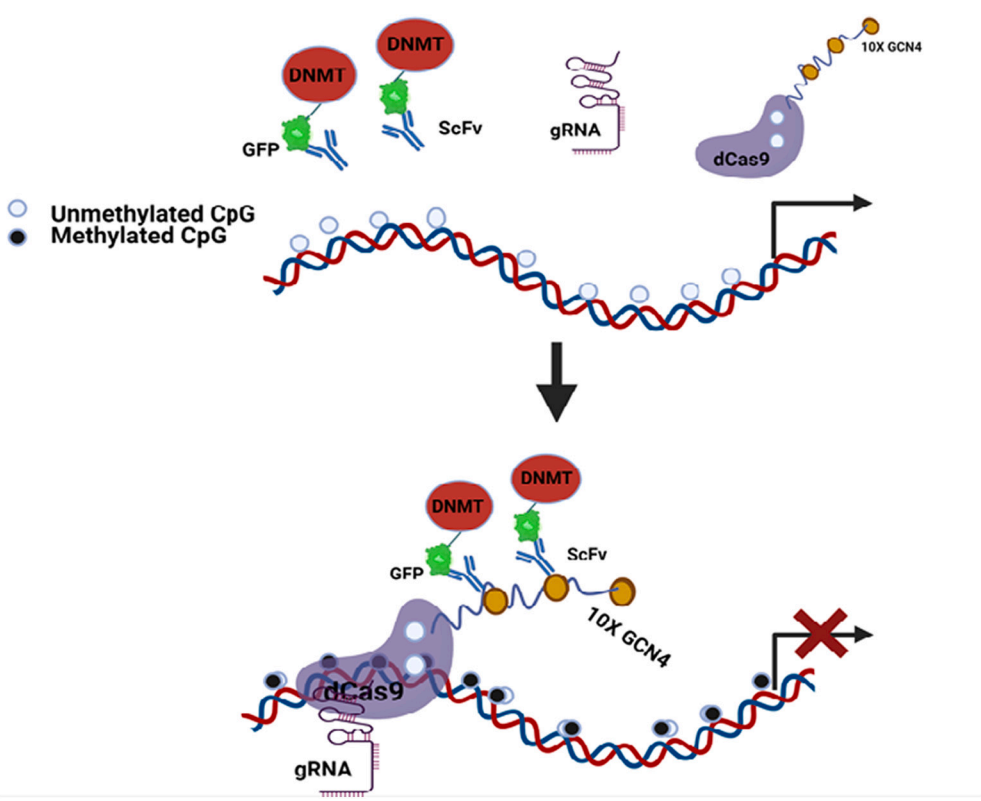

Fig. 2. DNMT or TET1 effectors tethered to dCas9 mediated epigenetic editing. DNA methylation occurs due to the collective effort of various components, i.e., gRNA/dCas9-effector complex. Here the effector includes DNMT, either attached to the dCas9 protein via a linker or repetitive peptide epitopes binding proteins, e.g., single-chain variable fragment (ScFv) antibody. gRNA to recruit the dCas9 nuclease to a targeted location on the genome. Here the effectors include DNMT, in which DNMTs add the methyl group to cytosine in DNA, resulting in a silencing effect.

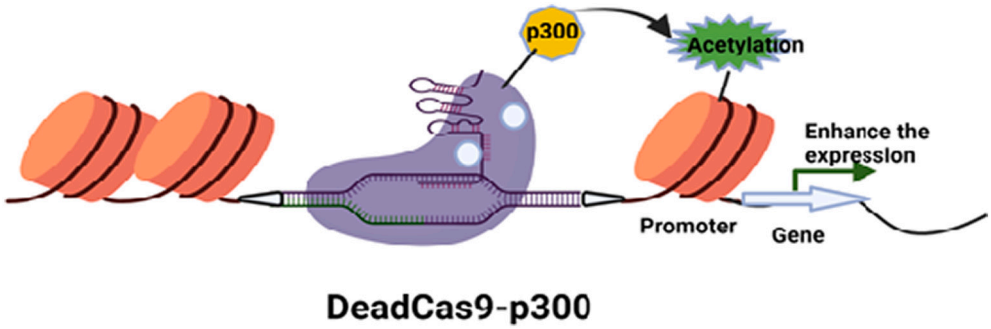

Fig. 3. dCas9-p300 chimeric protein enhances acetylation of H3K27 (histone H3 lysine 27) in the promoter region to increase gene expression.

demethylases (HDMs) cleave the methyl group from histone tails. Two vital enzymes [Jumonji C (JmjC) and Lysine-specific demethylase 1 (LSD1)] activate the demethylation of amino acid residues from histone tails (Fig. 4) [45].

4.5. CRISPR–dCas9 control by RNA switchable devices

The single guide RNA (sgRNA) binds to signal-responsive RNA aptamers and becomes a biosensor. The guided region of the single guide RNA pairs with the antisense stem in the absence of some specific signals; as a result, gRNA cannot bind to its target region. The guided part of sgRNA allows the conformational changes upon sensing clear indications by the RNA aptamers; therefore, sgRNA and dCas9 bind to the target DNA. The complex of sgRNA, RNA aptamers, and dCas9 trigger the modification of gene expression [133].

5. Application for epigenetic modification using CRISPR/Cas tools in plants

Recently developed dCas9-SunTag is a potential protein effector targeting system, which was employed as a transcriptional activator to understanding genomic functions and site-specific expression in *Arabidopsis* [129]. This system is used to activate targeted multiple loci like transposable elements and protein-coding genes. These new techniques are utilized for the site-specific manipulation of epigenomes in various plants. Gene function and gene transcription are controlled in *cis* through epigenetic modifications [129,134]. They observed that specific gene expression levels might directly or indirectly affect promoter methylation [129]. DNA methylation in the promoter region of gene activates the gene expression [133,134]. DNA demethylation with activation is used to understand the epigenetic regulation of a methylated loci [42]. Thus, the SunTag system is a potential technology for site-specific modification of the epigenetic status in plants and may have a broad range of advantages in the biotechnology fields [129]. The CRISPR/dCas9 system was a recently developed tool for gene function, gene deletions, and misexpression in host systems. It also enables the study of the gene regulation mechanism and understanding the multiple gene functions. The CRISPR/dCas9 offered to modify transcriptional and epigenetic marks in different plants [135]. The CRISPR/dCas9 systems are helpful to know the targeted gene regulation, activation or repression of transcriptional activation, and epigenetic modifications [128]. Lee et al. targeted the flowering time gene FLOWERING LOCUS T (FT) to induce epigenetic variations with different effector domains, including methyltransferase, a transcriptional activator, and acetyltransferase [128]. They observed alterations in flowering time. As a result, FT-modified transgenic lines showed modified phenotypes and flowering period with altered epigenetic regulation. A recently modified MS2-CRISPR/dCas9 system was developed to enhance the vigor of gene regulation. The Cas nuclease fused with the effector molecule binds to the target, and the effector molecule changes the epigenetic regulation in Arabidopsis, tobacco, and mammalian cells [128,136–138].

5.1. Transcriptional regulation in plants using CRISPRa/i

The fusion or recruitment of different transcriptional activators to the dCas9 became a routine method to target locus-specific transcriptional activation in plants [134]. The synthetic VP64 domain has been a commonly used activator in plants [139]. The other plant activators, such as VP64-p65-Rta and VPR, have been generated by combining VP64 with the activation domains of RelA NF-kappaB (p65) and Epstein–Barr virus Rta protein at the C-terminus [140]. The transcription factors, such as heat shock factor one (HSF1) and trans-activation domain (TAD) [137,141], belong to the plant ERF/EREBP family. The short motif of TAD is known as the 'EDLL' motif [141]. These transcriptional units such as EDLL peptide [139,140], VP64 [139], and the human HSF1 [137] have been employed successfully for targeted gene activation in several plant species [134]. Lowder et al. [139] used multiplexing strategies for transcriptional activation or repression in

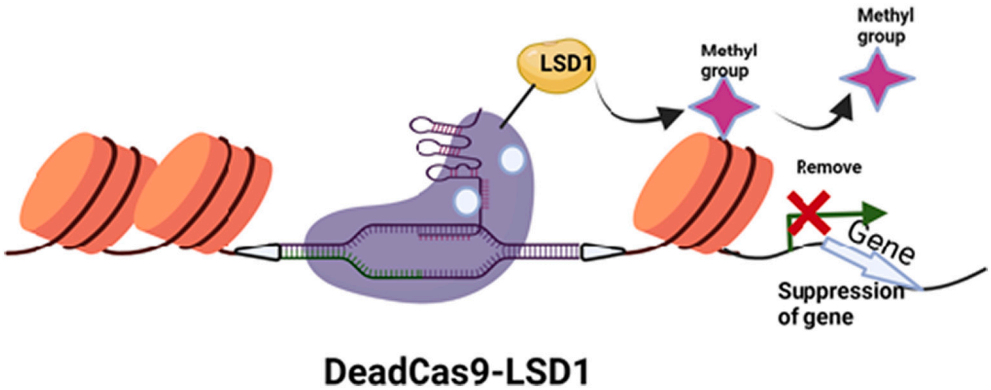

Fig. 4. dCas9-LSD1 removes methyl groups from the target site leads to the repression of the targeted gene.

Arabidopsis, rice, and tobacco. Several multiplex vector systems have been developed for transcriptional activation of endogenous genes by fusing the four tandem repeats of the transcriptional activator VP16 (VP64) with dCas9 [139]. Lowder et al. [142] developed the second generation of vector systems with multiplex transcription activator-like effector activation (mTALE-Act) for improved transcriptional activation in plants. This vector system (CRISPR-Act2.0) transcriptionally activated the four (two protein-coding and non-coding) genes in Arabidopsis and rice plants simultaneously [142].

The CRISPR/dCas9 system has been fused with different types of synthetic transcriptional repressors (CRISPRi) for targeted repression of various genes in plants [134]. Three endogenous loci (AtCSTF64, MiR159a, and MiR159b) were knockdown when the dCas9 fused with 3×-SRDX [139]. Vazquez-Vilar et al. [143] observed the efficient inhibition of transcription with a synthetic NOS promoter fused with luciferase by targeting the proximity of the transcription start site in tobacco (*Nicotiana benthamiana*). Tang et al. [144] observed the reduced mRNA levels (up to 90 %) of the endogenous MIR159b gene when the dCpf1–3×-SRDX had been fused with orthologous dCas9 system in Arabidopsis. The plant transcriptional co-repressor TOPLESS (a 300 amino acids N-terminus-N300TPL) fused with dCas9 inhibited the transcription of AtMED21 [145,146].

5.2. DNA methylation and demethylation in plants

A significant level of ectopic CHH methylation was observed in plant genome using NtDRM vector system fused with scFv [129]. The SunTag system and *Tobacco Rattle Virus* (TRV)-guide RNA were employed for both transcriptional activation and targeted DNA demethylation of the FWA gene. Heritable demethylation has been observed in Arabidopsis [147]. The SunTag vector system has been modified with dCas9 with catalytic domains of human demethylase TET1cd. The same vector system was successfully induced DNA demethylation at both FWA promoter and CACTA1 promoter loci in Arabidopsis [131]. The morphological and epigenetic modifications were observed when DNA methylation of promoters in two hypomethylated genes (AT3G17320 and AT5G28770) and two hypermethylated genes (AT1G72350 and AT1G09970) in Arabidopsis [135]. The fusion of the CRISPR/Cas9 based SunTag system with MQ1v more efficiently targets the DNA methylation of FWA promoter and observed early flowering phenotypes in Arabidopsis [147].

5.3. Histone acetylation and methylation in plants

The acetylation of N-terminal histone lysine eases DNA-histone binding and favors DNA accessibility and transcription activation [148]. The mammalian p300 histone acetyltransferase fused with dCas9 has been used to H3K27 acetylation in plants. This system efficiently increased the H3K27ac levels that target the FLOWERING TIME (FT) locus, a primary flowering time regulator. The H3K27ac slightly affected mRNA levels and flowering time [128,135]. The dCas9 was also fused with the Arabidopsis ubiquitous lysine acetyltransferase (AtHAT) to target the promoter of the AREB1 gene [149], which significantly improved drought stress tolerance in Arabidopsis [149,150]. The transcriptional outcome depends upon the methylation of histone lysine in plants. Different types of histone methyltransferases in plants contain the catalytic domain SET (Su(var), E(z), and Trithorax) domain. The FT locus, targeted via a dCas9/MS2 system H3K9 methyltransferase from Arabidopsis, was induced by the post-translational modification of histone [128].

Several abiotic stresses, including drought, heat, heavy metals, flooding, and salinity, severely hampered the growth and productivity of crop plants [132]. Park et al. [151] observed the transcriptional activation of two Arabidopsis genes [an endogenous Anthocyanin pigment1 (PAP1) and vacuolar H + -pyrophosphatase (AVP1)] significantly improved plant tolerance to drought stress. The high levels of Na + and K+ ions were accumulated in the transgenic plants of Arabidopsis expressing the AVP1 gene and exhibited tolerance to drought stress compared to the wild-type plants [151]. The *Arabidopsis histone acetyltransferase1* (*AtHAT1*) fused with CRISPR/dCas9 was also employed to trigger the abscisic acid (ABA)-responsive element binding protein1/ABRE binding factor (AREB1/ABF2). *AtHAT1* enhanced the drought resistance in the transgenic Arabidopsis plants, which showed higher chlorophyll content and survival rate than the control plants [149]. The transgenic Arabidopsis plants with AREB-1 activated by CRISPRa showed enhanced drought tolerance compared with their wild-type plants, evidenced by a low malonaldehyde level and a two-fold increase in relative water content levels [150].

6. Future prospective

Epigenetic regulation plays a crucial role in plant development. Research about epigenetics in plants has advantages as compared with animals. In plants, there is no need to maintain germline, and plants enable the transfer of epigenetic information throughout gametogenesis [151–154]. However, epigenetic marks may be unstable in plants. We need to focus on stable and heritable epigenetic mechanisms over generations, and we need to know where epigenetic states can be stably transmitted to their progeny [152]. Climatic stress conditions affect crop yield and quality and cause economic loss. The majority of the epigenetic mechanism studies were carried out in Arabidopsis, which has led to an interest in plants' epigenetic studies to better adapt to climatic changes and the breeding of climate-smart crops [153]. A better understanding of epigenetic marks helps plant breeders produce new breeding tools by exploiting phenotypic variation in existing crop plants. Nowadays, various techniques are currently available to alter the targeted site to get epigenetic modifications. Among these techniques, the newly developed CRISPR/Cas9 based tools open a new era for epigenetic modification and regulation. CRISPR/Cas9 genome editing is a robust and Nobel-prize-winning precise genome editing technology, which has shown the versatile power of editing specific genome sequences both genetically and epigenetically [18,155,156]. The DNA binding domain, such as dCas9, ZFNs, and TALENs nucleases, are used to activate or repress the chromatin mark at particular loci and affect the plant phenotype during plant development and stress exposure [157]. The epigenetic single nucleotide polymorphisms were used to alter the chosen gene and exploitation of epigenetic variation in crops that could lead to the development of a new transgene-free breeding method [160]. Thus, all these tools enable the creation of unique varieties of the smart climatic crop. The appropriate usage of these tools allows for a better understanding of epigenetic traits, alteration in DNA methylation and histone modification, and plant inheritance patterns. In the short future, researchers need to pay attention to epigenetic modification and its role in adapting to climatic changes to develop new stress-resistant crops. However, more research is needed to achieve the goals of controlling epigenetics in the plant for both functional study and crop improvement.

6.1. Discovering new Cas and modified Cas nucleases will enhance the application of CRISPR/Cas on epigenetic fields

Although a great process has been made in discovering new Cas and modified current Cas nucleases, PAM requirements are still one of the major limitations of the CRISPR/Cas system, which limits the application of CRISPR/Cas tools for editing all potential gene sequences genetically and epigenetically. Continually identifying new Cas and modified current Cas nucleases, particularly for PAM-less required Cas enzymes, will expand the usage of CRISPR/Cas tools [158,159]. It is also possible in the future that a PAM-independent CRISPR/Cas system will be developed for editing an individual genome sequence.

6.2. Off-target impact is also a major issue for epigenome editing

CRISPR/Cas genome editing is genetic, so the off-target impact is also a big problem for CRISPR epigenetic editing. Reducing or even eliminating off-target impacts will enhance the application of CRISPR/Cas tools. There are many potential methods that can be employed to achieve these goals, including designing high affinity sgRNA and using the right Cas enzymes. Selecting the right CRISPR/Cas regent delivery system also can reduce the off-target impact. The *Agrobacterium*-mediated gene transformation is the major approach to delivering both Cas protein gene and sgRNA gene into plant cells; however, this method requires both Cas and sgRNA genes integrated into the plant genome, which keeps the CRISPR/Cas system in the plant cell for a long period and potentially causes continuous genome editing, including off-target editing. The delivery of a Cas ribonucleoprotein (RNP) complex only allows the CRISPR/Cas regents that existed in plant cell for a limited time, which may reducing the off-target impacts [160–162].

6.3. Designing new CRISPR/Cas epigenetic tools for expanding epigenetic editing

Both CRISPR and epigenetic editing are two new fields; although there are many computational tools and resources have been developed for CRISPR/Cas-related research and application [163], there are still lots of topics waiting to explore for facing the quick development of this field. Any enzymes and biological processes relating to epigenetic regulation can be linked to the CRISPR/Cas systems by fusing the enzymes with dCas9 and/or nCas9. Thus, exploring new enzymes regulating epigenetic regulation will help to develop new epigenetic CRISPR tools.

7. Conclusion

In the past decades, epigenetic inheritance has been attracting more and more attention from both scientific and industrial communities due to its unique characteristics and function that it alters plant traits but not causes DNA sequence change. The most common epigenetic changes include chromatin remodeling, histone modification, DNA methylation and modification, and non-coding RNAs. Thus, it is a new strategy for studying gene function and improving plant by using epigenetic engineering. However, the lack of reliable tools for manipulating epigenetic change hinders the application of epigenetics. The most advanced CRISPR/Cas genome editing has opened a new era for epigenetics-related study and application. Cas nuclease has two independent functions: binding and cutting DNA sequences at a precise location; deactivating Cas cutting function does not affect its binding function. More importantly, Cas9 can be fused with another functional domain, such as an epigenetic modifier, e.g. a transcriptional activator or repressor domain, or a DNA methyltransferase (DNMT) or demethylase (TET) domain, to form a new epigentic tool for manipulating epigenetic change in plants. Currently, it has been becoming a new research hotspot to develop CRISPR/Cas epigentic editors and to use these newly developed CRISPR/Cas epigentic editors for enhancing plant growth, development and tolerance to various environmental stresses as well as gene functional stdies.

Declaration of competing interest

The authors declare no conflict of interest.

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